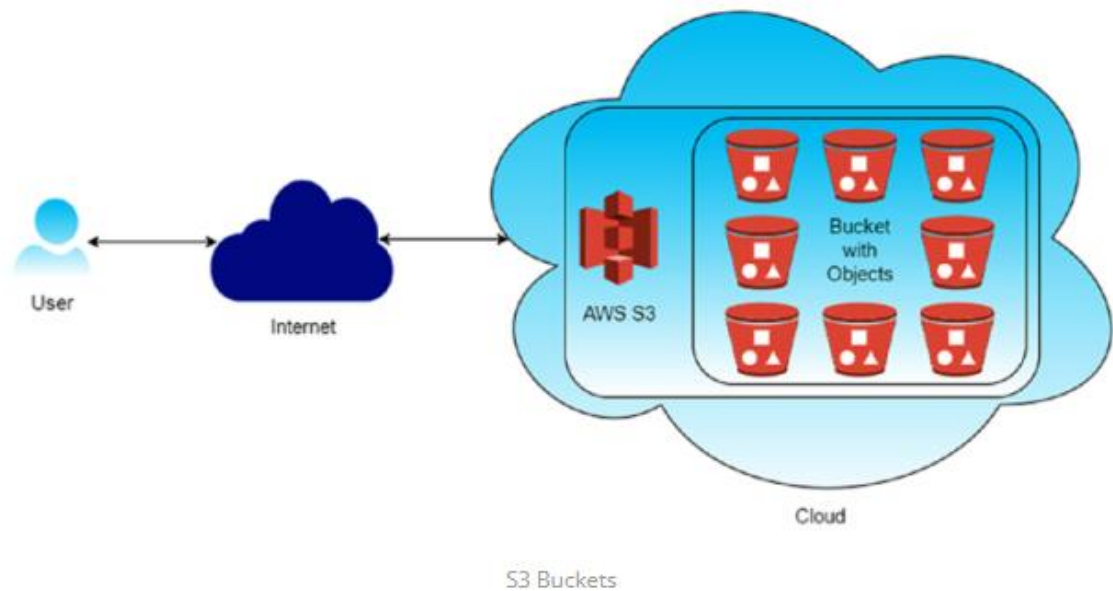
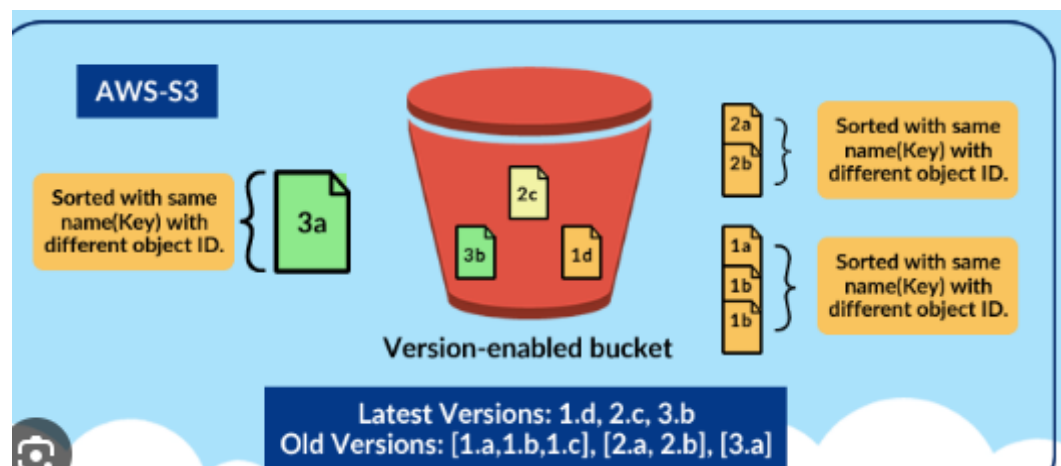


Week-4 Simple Storage Service (S3)

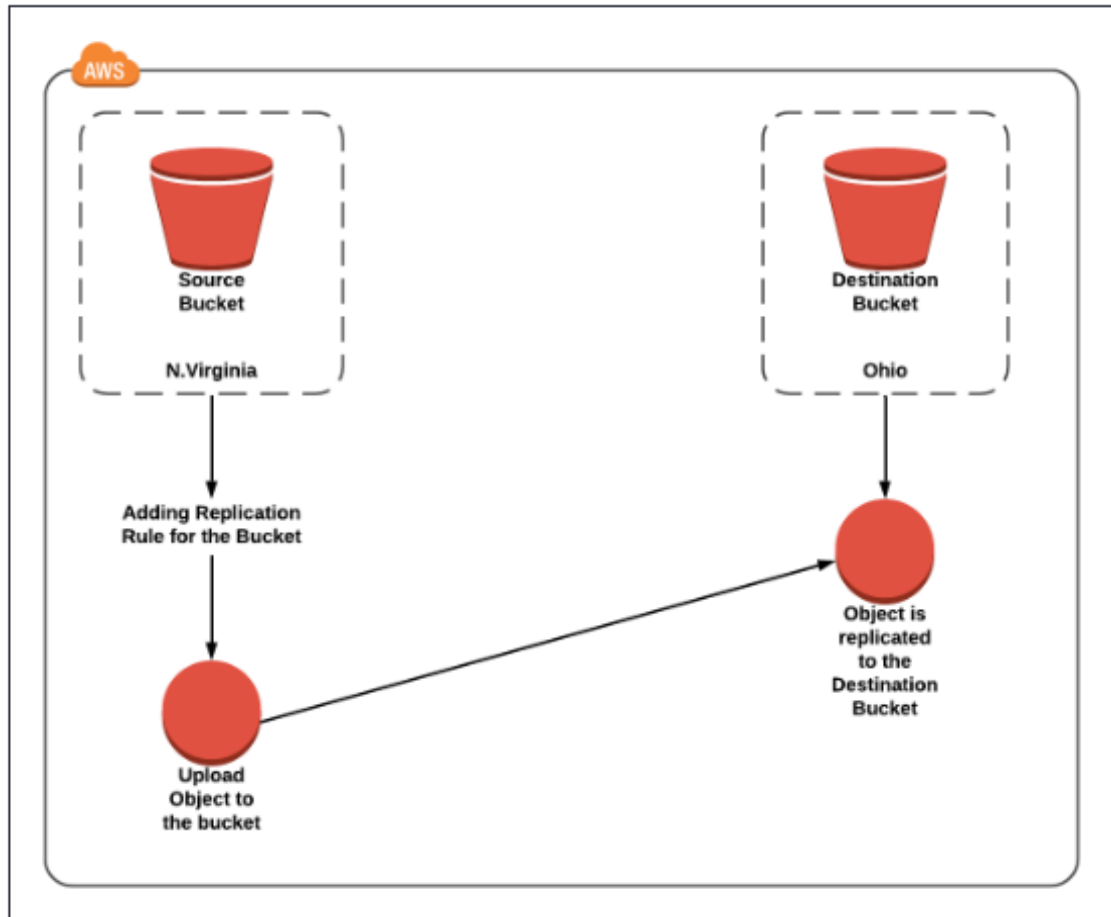


1. Versioning: Versioning in Amazon S3 is a feature that allows you to keep multiple versions of an object in the same bucket. Whenever you upload a new version of an object with the same key (name) as an existing object, S3 doesn't overwrite the existing object but instead creates a new version of it. This means you can retain and access previous versions of objects, providing an additional layer of protection against accidental deletions or overwrites

- i. **Accidental Deletion Protection** :: With versioning enabled, if you accidentally delete an object, you can still retrieve previous versions of it, preventing data loss.
- ii. **Accidental Overwrite Protection** :: Versioning prevents accidental overwrites of objects. Even if you upload a new version of an object with the same name, the previous versions are preserved.



1. **Cross-Region Replication:** Cross-region replication (CRR) in Amazon S3 is a feature that automatically replicates objects from one bucket in one AWS region to another bucket in a different AWS region. This provides redundancy and disaster recovery capabilities, ensuring that your data remains available even if an entire AWS region becomes unavailable.
 - i. **Disaster Recovery:** Cross-region replication helps ensure business continuity by replicating critical data to a different geographic region, reducing the risk of data loss due to regional disasters or outages.
 - ii. **Low-Latency Access:** Users in different geographic regions can access data from the nearest region, reducing latency and improving performance.
 - iii. **Compliance:** Some regulatory requirements mandate data replication across multiple geographic regions for data sovereignty and compliance reasons.
2. **Event Notifications:** S3 can trigger events (e.g., object creation, deletion) that can be routed to AWS Lambda, SNS, or SQS for processing.
3. **Integration:** S3 seamlessly integrates with other AWS services such as AWS Lambda, Amazon CloudFront, AWS Glue, Amazon Athena, and Amazon EMR, enabling you to build powerful data processing and analytics workflows.

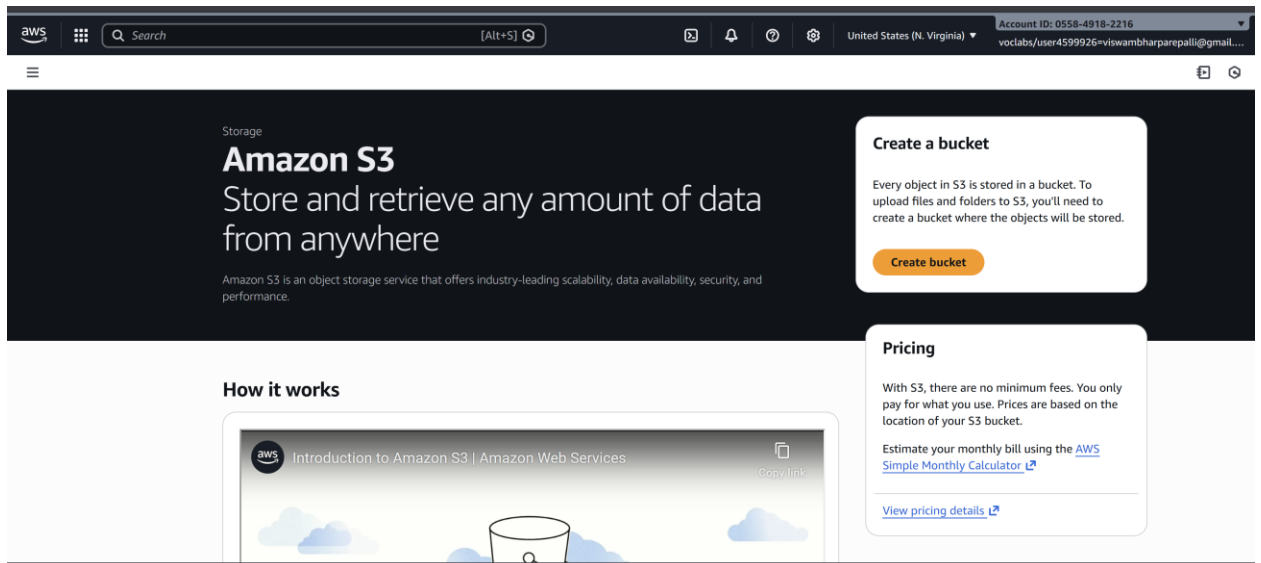


1. Creating an S3 bucket, upload an object, and enable public access

☒ Create an S3 Bucket (with Public Access Enabled)

Step 1: Open S3 Service

- Login to AWS Management Console
- Go to Services → S3



Storage

Amazon S3

Store and retrieve any amount of data from anywhere

Amazon S3 is an object storage service that offers industry-leading scalability, data availability, security, and performance.

Create a bucket

Every object in S3 is stored in a bucket. To upload files and folders to S3, you'll need to create a bucket where the objects will be stored.

[Create bucket](#)

How it works

Introduction to Amazon S3 | Amazon Web Services

[Copy link](#)

Pricing

With S3, there are no minimum fees. You only pay for what you use. Prices are based on the location of your S3 bucket.

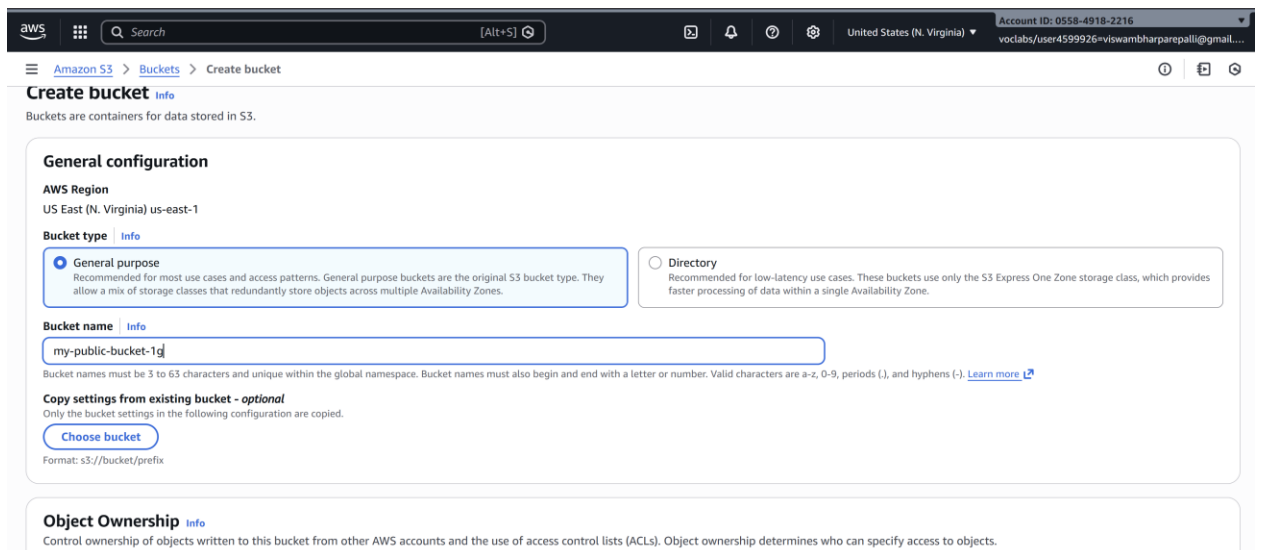
Estimate your monthly bill using the [AWS Simple Monthly Calculator](#)

[View pricing details](#)

-
- Click Create bucket

Step 2: Bucket Configuration

- Bucket name: my-public-bucket-123
- Region: Choose nearest region



Amazon S3 > Buckets > Create bucket

Create bucket

Buckets are containers for data stored in S3.

General configuration

AWS Region
US East (N. Virginia) us-east-1

Bucket type

☒ **General purpose**
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ **Directory**
Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name
my-public-bucket-123

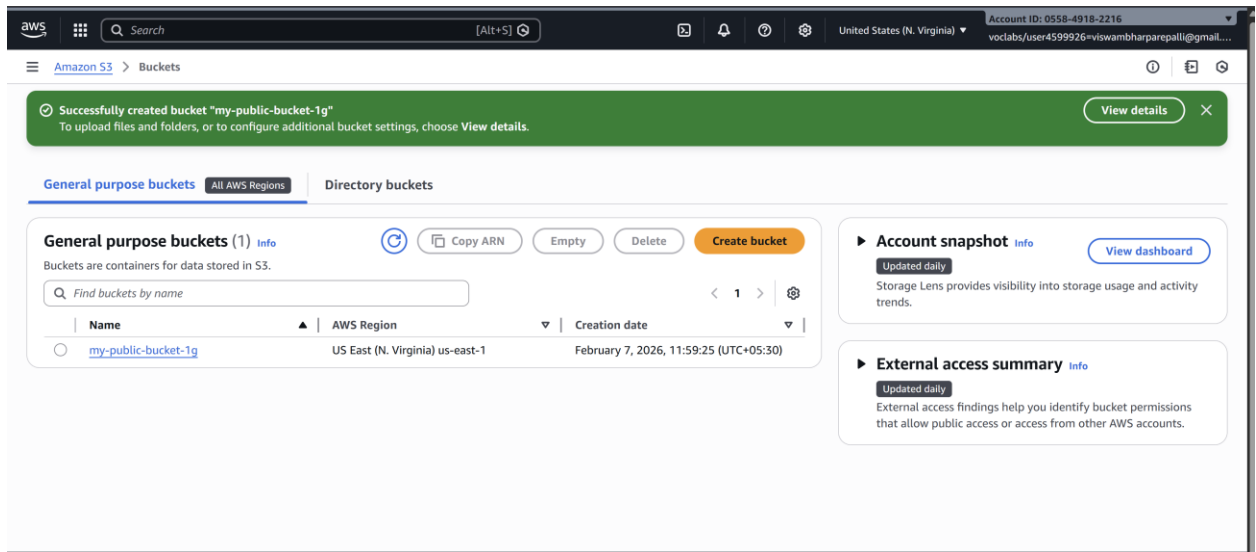
Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional
Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership
Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.



Step 3: Disable Block Public Access (IMPORTANT)

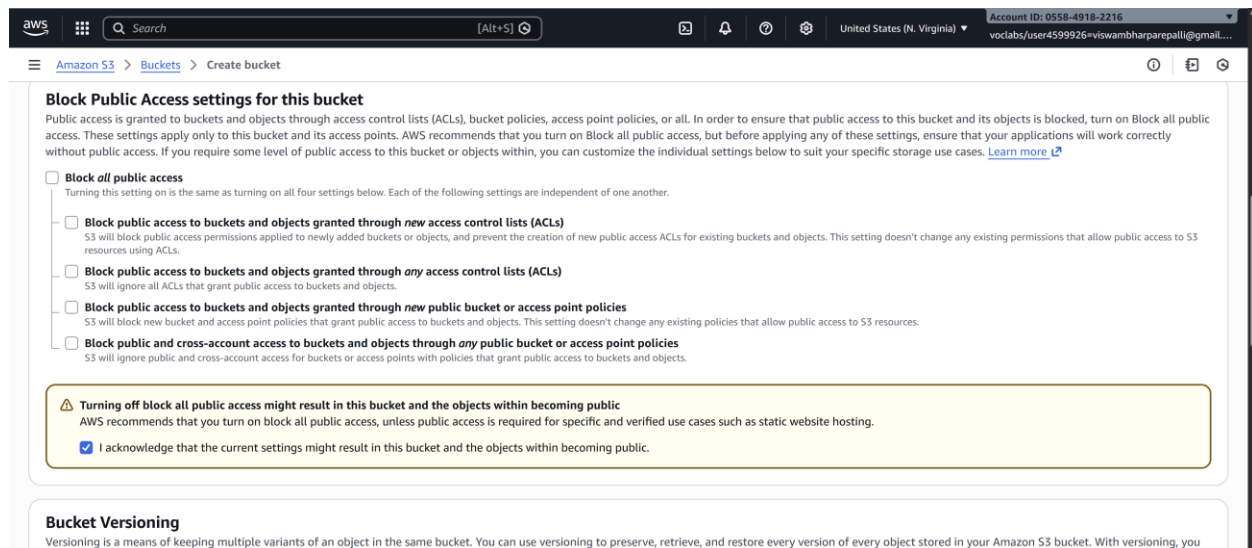
Under Block Public Access settings for this bucket:

✗ Uncheck Block all public access

You must uncheck ALL options:

✓ Check I acknowledge that this bucket will become public

👉 Click Create bucket



2 Enable Public Access at Bucket Level (ACL)

Step 4: Open the Bucket

- Click on the bucket name

- Go to Permissions tab

The screenshot shows the AWS IAM console interface. At the top, there's a navigation bar with the AWS logo, a search bar, and account information (United States (N. Virginia), Account ID: 0558-4918-2216). Below the navigation bar, the breadcrumb trail shows 'Amazon S3 > Buckets > my-public-bucket-1g'. The main content area has tabs for 'Objects', 'Metadata', 'Properties', 'Permissions' (selected), 'Metrics', 'Management', and 'Access Points'. Under the 'Permissions' tab, there are three sections: 'Permissions overview' with an 'Access finding' link, 'Block public access (bucket settings)' with an 'Edit' button, and 'Bucket policy' with 'Edit' and 'Delete' buttons. The 'Block public access' section shows a warning icon and the text 'Block all public access' with a toggle set to 'Off'.

Step 5: Edit Bucket ACL

- Scroll to Access Control List (ACL)
- Click Edit

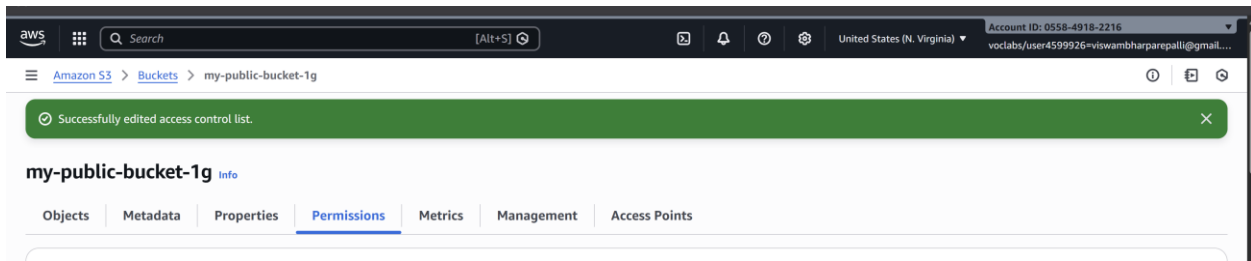
Under Public access:

- Everyone (public access) → ✓ Read access

✓ Save changes

The screenshot shows the 'Edit access control list (ACL)' page in the AWS IAM console. The breadcrumb trail is 'Amazon S3 > Buckets > my-public-bucket-1g > Edit access control list (ACL)'. The page has a table with three columns: 'Grantee', 'Objects', and 'Bucket ACL'. The 'Grantee' column lists four groups: 'Bucket owner (your AWS account)', 'Everyone (public access)', 'Authenticated users group (anyone with an AWS account)', and 'S3 log delivery group'. The 'Objects' column has checkboxes for 'List' and 'Write' permissions. The 'Bucket ACL' column has checkboxes for 'Read' and 'Write' permissions. The 'Everyone (public access)' group has 'Read' checked and 'Write' disabled. A warning box at the bottom states: 'When you grant access to the Everyone or Authenticated users group grantees, anyone in the world can access the objects in this bucket. I understand the effects of these changes on my objects and buckets.' There is also a section for 'Access for other AWS accounts' which shows 'No other AWS accounts associated with the resource.'

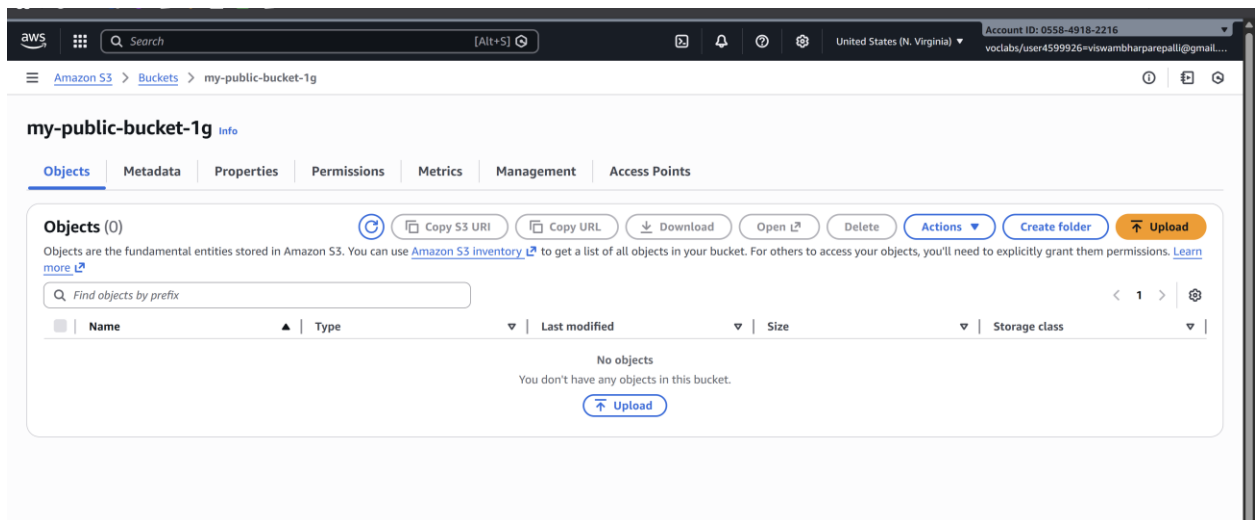
This allows public access at bucket level



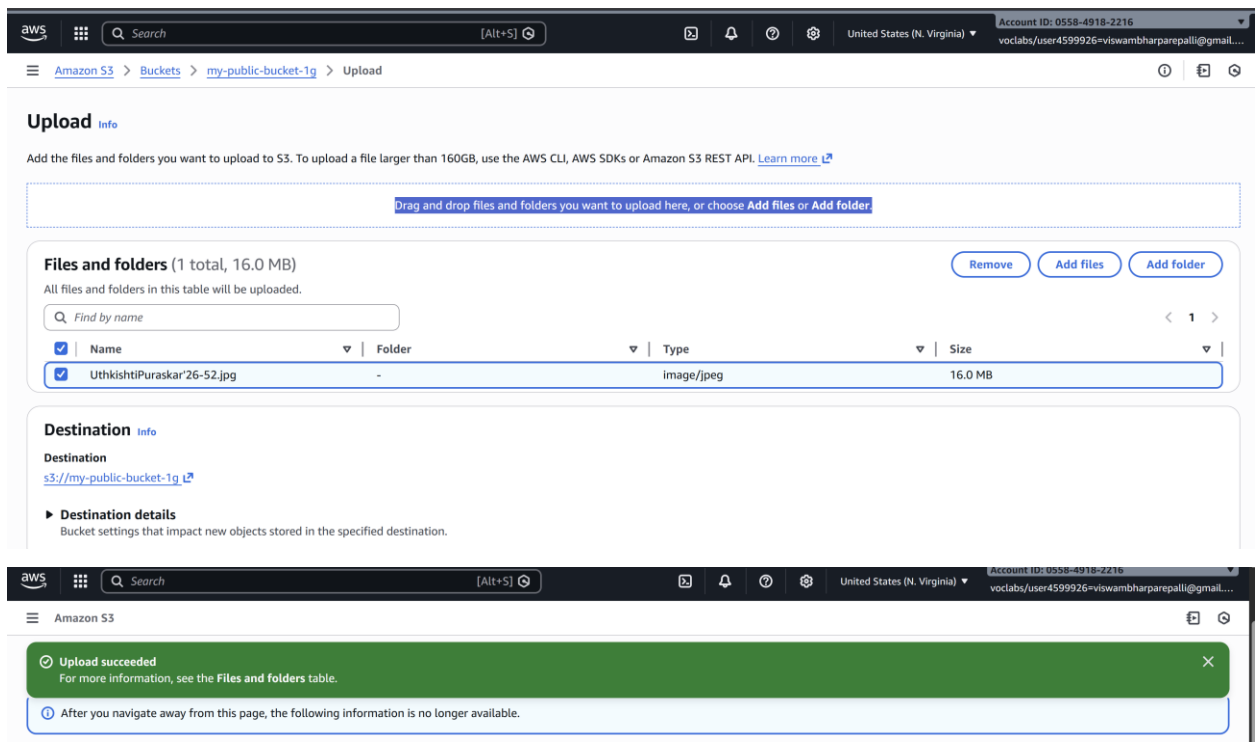
8 Upload Object to the Bucket

Step 6: Upload File

- Go to Objects tab
- Click Upload



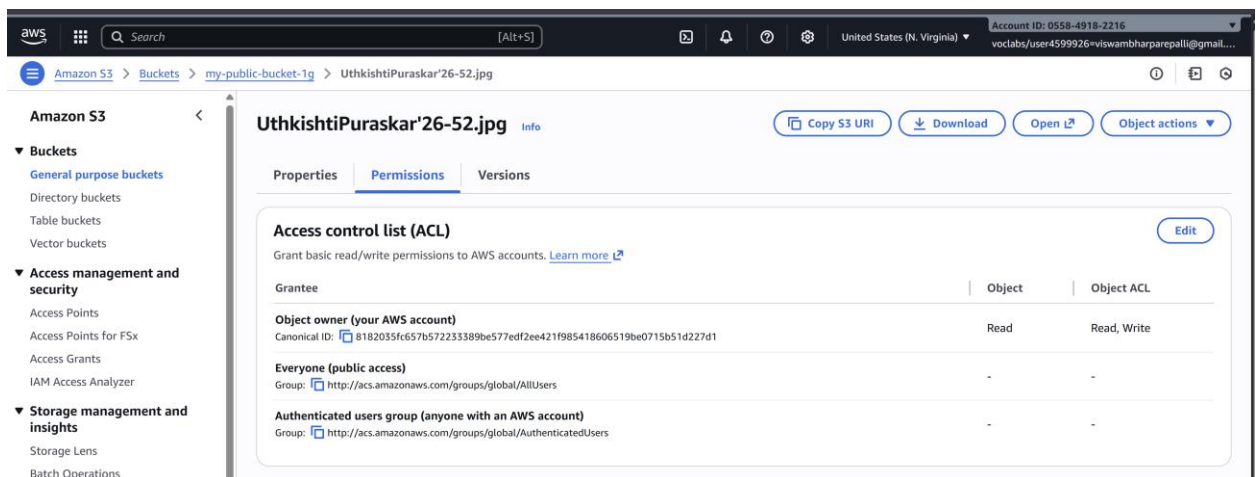
-
- Click Add files
- Select file (e.g., image.jpg)
- Click Upload



4 Enable Public Access for Object (ACL)

Step 7: Object-Level ACL Permission

- Click on uploaded object
- Go to Permissions tab



- Scroll to Access Control List (ACL)
- Click Edit

Under Public access:

- Everyone (public access) → ✓ Read access

aws Search [Alt+S] United States (N. Virginia) Account ID: 0558-4918-2216 voclabs/user4599926-viswambharparepalli@gmail...

Amazon S3 > Buckets > my-public-bucket-1g > UthkishtiPuraskar'26-52.jpg > Edit access control list

Edit access control list Info

Access control list (ACL)
Grant basic read/write permissions to AWS accounts. [Learn more](#)

Grantee	Objects	Object ACL
Object owner (your AWS account) Canonical ID: 8182035fc657b572233389be577edf2ee421f985418606519be0715b51d227d1	☒ Read	☒ Read ☒ Write
Everyone (public access) Group: http://acs.amazonaws.com/groups/global/AllUsers	☒ ☒ Read	☒ ☒ Read ☐ Write
Authenticated users group (anyone with an AWS account) Group: http://acs.amazonaws.com/groups/global/AuthenticatedUsers	☐ Read	☐ Read ☐ Write

⚠ When you grant access to the Everyone or Authenticated users group grantees, anyone in the world can access this object. [Learn more](#)

☒ I understand the effects of these changes on this object.

• Access for other AWS accounts

✓ Save changes

aws Search [Alt+S] United States (N. Virginia) Account ID: 0558-4918-2216 voclabs/user4599926-viswambharparepalli@gmail...

Amazon S3 > Buckets > my-public-bucket-1g > UthkishtiPuraskar'26-52.jpg

Successfully edited access control list for object "UthkishtiPuraskar'26-52.jpg".

UthkishtiPuraskar'26-52.jpg Info

[Copy S3 URI](#) [Download](#) [Open](#) [Object actions](#)

Verify Public Access

Step 8: Test Object URL

- Copy Object URL

Account ID: 0558-4918-2216
voclabs/user4599926=viswambharpaparepalli@gmail....

United States (N. Virginia) ▼

Object actions

Copy S3 URI Download Open ↗

S3 URI
s3://my-public-bucket-1g/UthkishtiPuraskar'26-52.jpg

Amazon Resource Name (ARN)
arn:aws:s3:::my-public-bucket-1g/UthkishtiPuraskar'26-52.jpg

Entity tag (Etag)
ba27746a58d09db5808678

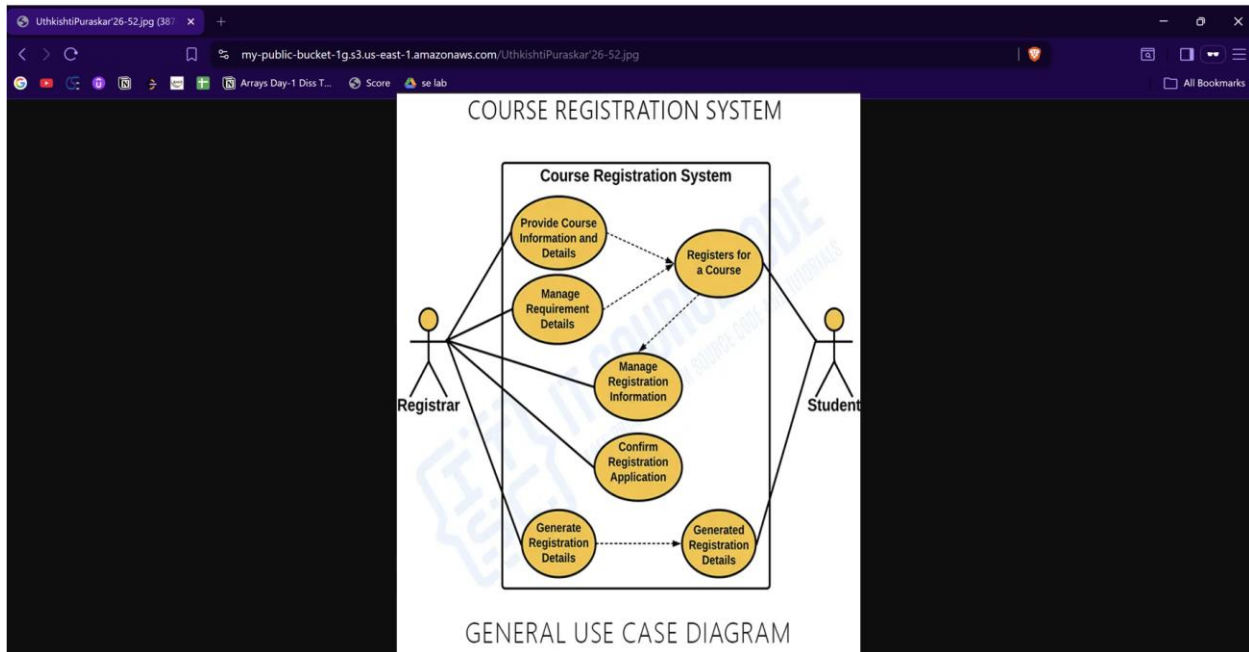
Object URL Copied

Object URL
https://my-public-bucket-1g.s3.us-east-1.amazonaws.com/UthkishtiPuraskar'26-52.jpg

-

- Open in browser (incognito)

✓ Object should be accessible without login



2. S3 Versioning & Cross-Region Replication

1. Enabling Versioning in S3 (with Output Scenario)

◆ Objective

To enable version control on an S3 bucket and observe object versions.

◆ Steps to Enable Versioning

1. Open AWS Management Console
2. Navigate to S3
3. Select the bucket
4. Go to Properties tab
5. Scroll to Bucket Versioning

6. [Edit](#)

7. Click Edit

8. Select Enable

9. Click Save changes

✓ Versioning is now enabled

After enabling Bucket Versioning, you might need to update your lifecycle rules to manage previous versions of objects.

Cancel Save changes

Successfully edited Bucket Versioning
To transition, archive, or delete older object versions, [configure lifecycle rules](#) for this bucket.

◆ Output Scenario (What You Observe)

Step A: Upload Object

- Upload file: report.pdf

✂ Output:

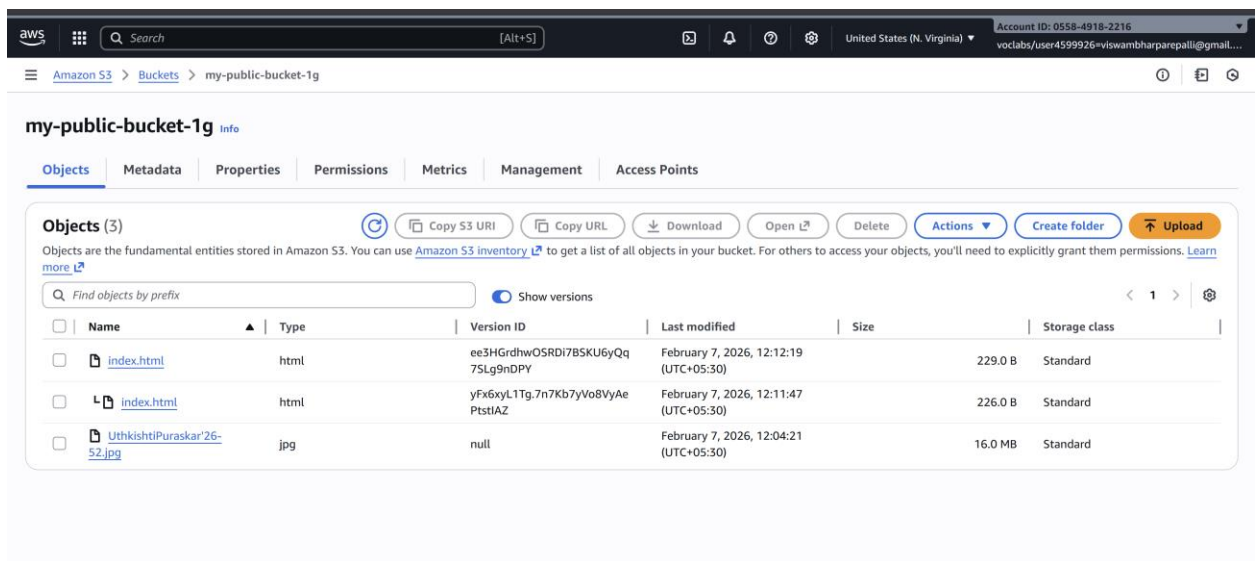
- Version ID is assigned automatically

Step B: Modify and Re-upload Same Object

- Upload report.pdf again (same name)

✂ Output:

- Old version is preserved
- New version becomes current



The screenshot shows the AWS Management Console interface for a bucket named 'my-public-bucket-1g'. The 'Objects' tab is selected, displaying a list of objects. The table shows two versions of 'index.html' and one object named 'UthkishtiPuraskar'26-52.jpg'.

	Name	Type	Version ID	Last modified	Size	Storage class
☐	index.html	html	ee3HGdhwOSRD17BSKU6yQq75Lg9nDPY	February 7, 2026, 12:12:19 (UTC+05:30)	229.0 B	Standard
☐	index.html	html	yFx6xyl1Tg.7n7Kb7yVo8VyAePtstIAZ	February 7, 2026, 12:11:47 (UTC+05:30)	226.0 B	Standard
☐	UthkishtiPuraskar'26-52.jpg	jpg	null	February 7, 2026, 12:04:21 (UTC+05:30)	16.0 MB	Standard

Step C: Delete Object

- Click Delete on report.pdf

Delete objects [Info](#)

⚠️ If a folder is selected for deletion, all objects in the folder will be deleted, and any new objects added while the delete action is in progress might also be deleted. If an object is selected for deletion, any new objects with the same name that are uploaded before the delete action is completed will also be deleted.

- Deleting the specified objects can't be undone.

[Learn more](#)

Specified objects

Find objects by name

Name	Version ID	Type	Last modified	Size
index.html	yFx6xyl1Tg.7n7Kb7yVo8VyAePtstIAZ	html	February 7, 2026, 12:11:47 (UTC+05:30)	226.0 B

Permanently delete objects?

To confirm deletion, type *permanently delete* in the text input field.

permanently delete

[Cancel](#) [Delete objects](#)

✂️ Output:

- Object appears deleted
- A Delete Marker is created
- Older versions still exist and can be restored

Successfully deleted objects
View details below.

ⓘ After you navigate away from this page, the following information is no longer available.

Summary

Source	Successfully deleted	Failed to delete
s3://my-public-bucket-1g	✓ 1 object, 226.0 B	0 objects

[Failed to delete](#) [Configuration](#)

🔄 Cross-Region Replication (CRR)

◆ Objective

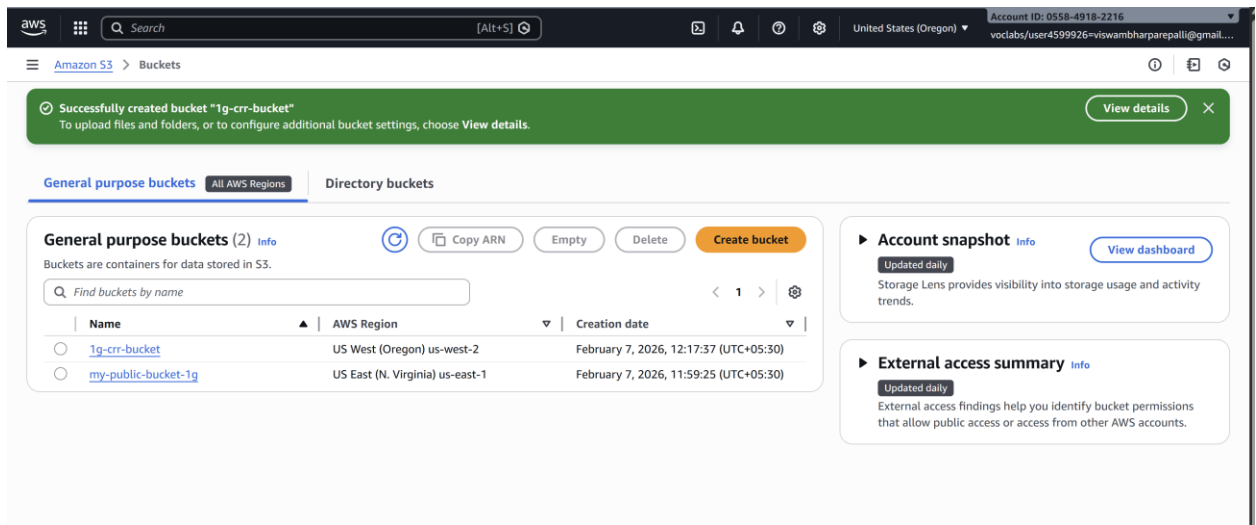
To replicate objects from one bucket to another in a different region using AWS-managed permissions.

◆ Prerequisites (Very Important)

- ✓ Versioning must be enabled on both buckets
- ✓ Source and destination buckets must be in different regions
- ✓ AWS Academy automatically handles replication permissions

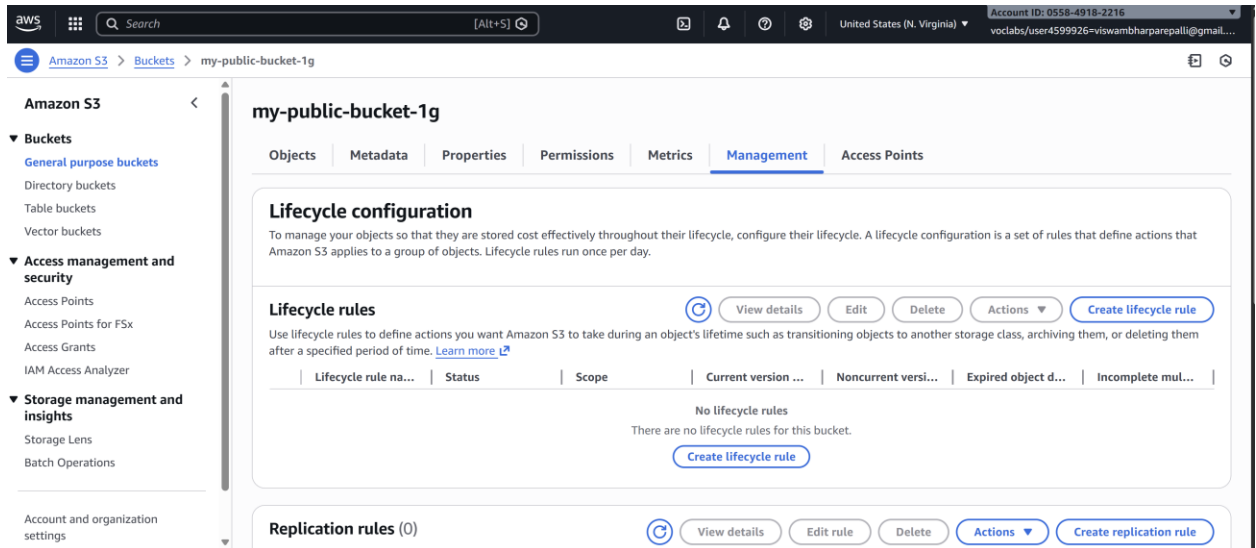
Step 1 Create Destination Bucket

1. Go to S3 → Create bucket
2. Bucket name: dest-crr-bucket
3. Select different region
4. Complete bucket creation
5. Enable Versioning (same steps as above)

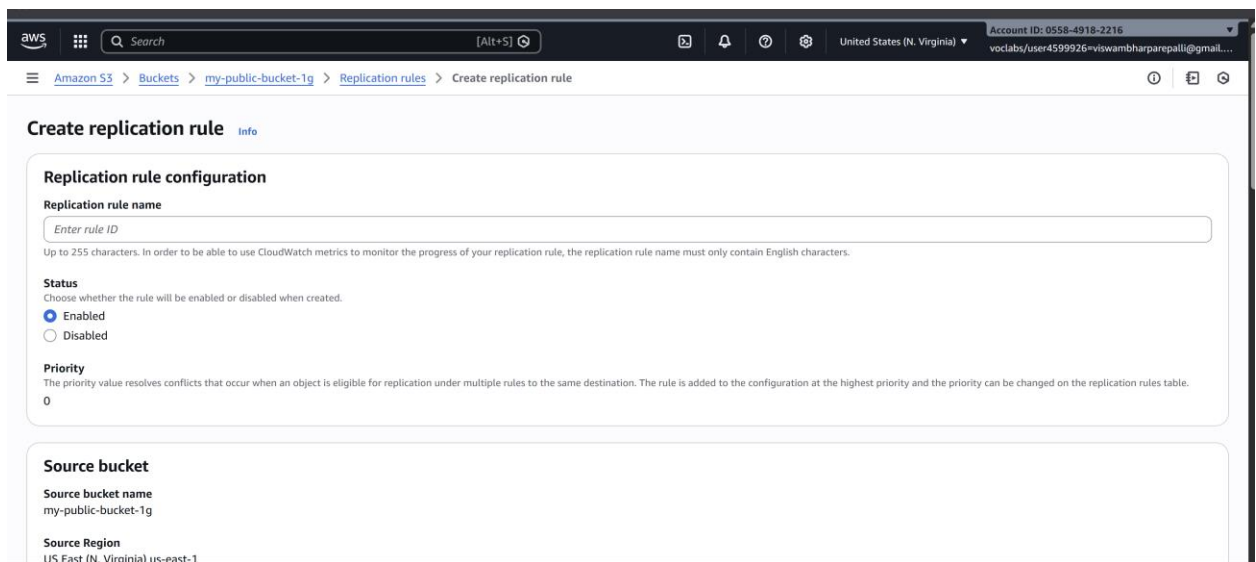


Step 2 Create Replication Rule (Without IAM Selection)

1. Open Source bucket
2. Go to Management tab



- 3.
4. Click Replication rules
5. Click Create replication rule



- 6.

Step 1: Configure Replication Rule

- Rule name: academy-crr-rule
- Status: Enabled
- Scope: Apply to all objects

Account ID: 0558-4918-2216
voclabs/user4599926=viwambharpaparepalli@gmail...

United States (N. Virginia)

Amazon S3 > Buckets > my-public-bucket-1g > Replication rules > Create replication rule

Create replication rule [Info](#)

Replication rule configuration

Replication rule name

academy-crr-rule

Up to 255 characters. In order to be able to use CloudWatch metrics to monitor the progress of your replication rule, the replication rule name must only contain English characters.

Status

Choose whether the rule will be enabled or disabled when created.

☒ Enabled

☐ Disabled

Priority

The priority value resolves conflicts that occur when an object is eligible for replication under multiple rules to the same destination. The rule is added to the configuration at the highest priority and the priority can be changed on the replication rules table.

0

Source bucket

Source bucket name

my-public-bucket-1g

Source Region

US East (N. Virginia) us-east-1

Choose a rule scope

- ☐ Limit the scope of this rule using one or more filters
- ☒ Apply to all objects in the bucket

Step 4 Choose Destination Bucket

- Select Another bucket
- Choose Destination bucket
- Confirm different region

AWS Academy Note:

You will see an option like:

"AWS will create and manage the required permissions automatically"

✓ Select Use AWS-managed role / default role

Destination

You can replicate objects across buckets in different AWS Regions (Cross-Region Replication) or you can replicate objects across buckets in the same AWS Region (Same-Region Replication). You can also specify a different bucket for each rule in the configuration. [Learn more](#) or see [Amazon S3 pricing](#)

☒ Choose a bucket in this account
☐ Specify a bucket in another account

Bucket name
 Choose the bucket that will receive replicated objects.

1g-crr-bucket [Browse S3](#)

Destination Region
 US West (Oregon) us-west-2

Step 5 Replication Options

- **Replicate:**
 - ✓ New objects
 - ✗ Existing objects (optional, depends on Academy permissions)
- **Encryption: Leave default**

Click Save

✓ Replication rule is now active

Replicate existing objects?

You can enable a one-time Batch Operations job from this replication configuration to replicate objects that already exist in the bucket and to synchronize the source and destination buckets. [Learn more](#) or [see pricing](#)

Existing objects

☒ No, do not replicate existing objects.
☐ Yes, replicate existing objects.

[Cancel](#) [Submit](#)

Replication configuration settings

Configuration settings affect all replication rules in the bucket.

Source bucket
 my-public-bucket-1g

Source Region
 US East (N. Virginia) us-east-1

Replication rules (1)

Replication rule name	Status	Destination bucket	Destination Region	Priority	Scope	Storage class	Replica owner	Replication Time Control	KMS-encrypted objects (SSE-KMS or DSSE-MANAGED)	Replica modification sync

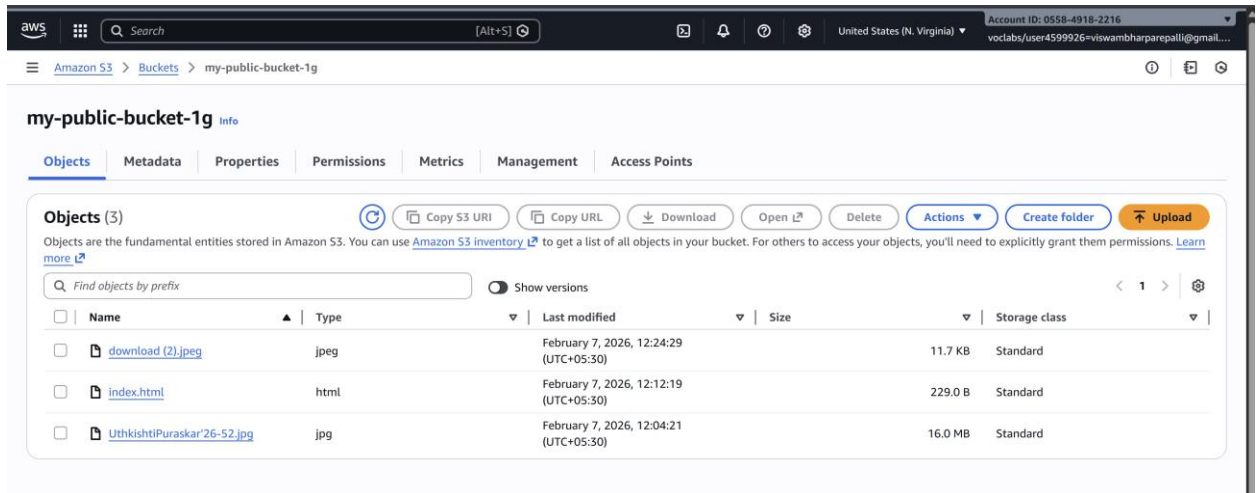
3.1 CRR Output Scenario (What You Observe)

◆ Step A: Upload New Object to Source Bucket

- Upload file: image.png

✂ Output:

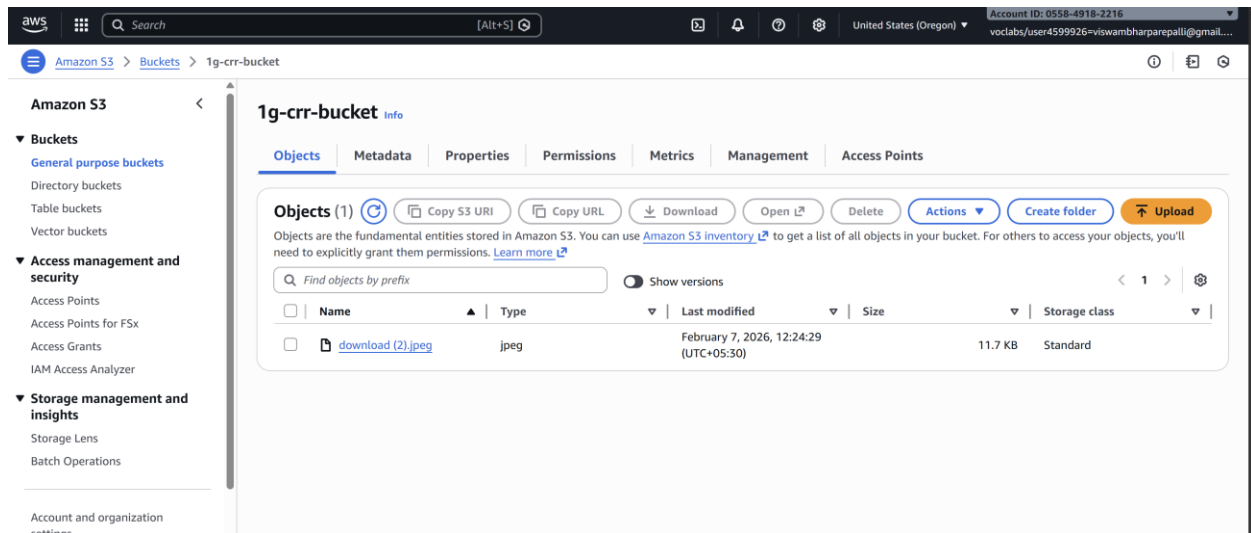
- Object is uploaded normally



◆ Step B: Verify Destination Bucket

- Open destination bucket
- Object image.png appears automatically
- Version ID is present

✓ Replication successful



3. Static Website Hosting in S3

(Using Bucket-Level & Object-Level Permissions – ACL Method)

◆ Objective

To host a static website in Amazon S3 using index.html and error.html, and allow public access using ACL-based bucket and object permissions.

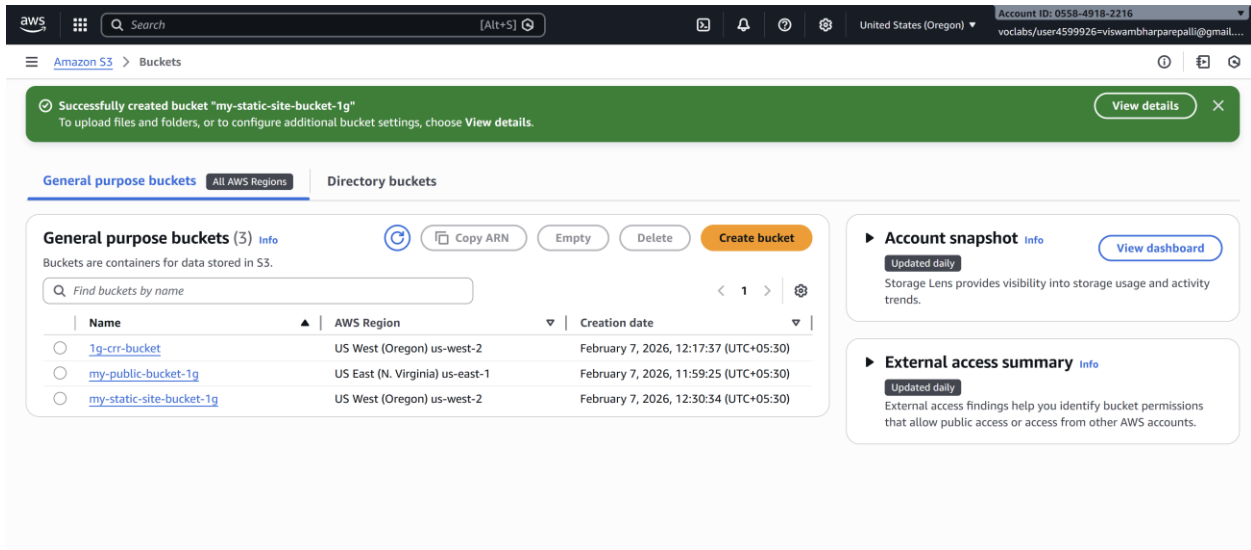
◆ Prerequisites

- AWS / AWS Academy account
 - Two files:
 - index.html
 - error.html
-

📁 Create an S3 Bucket

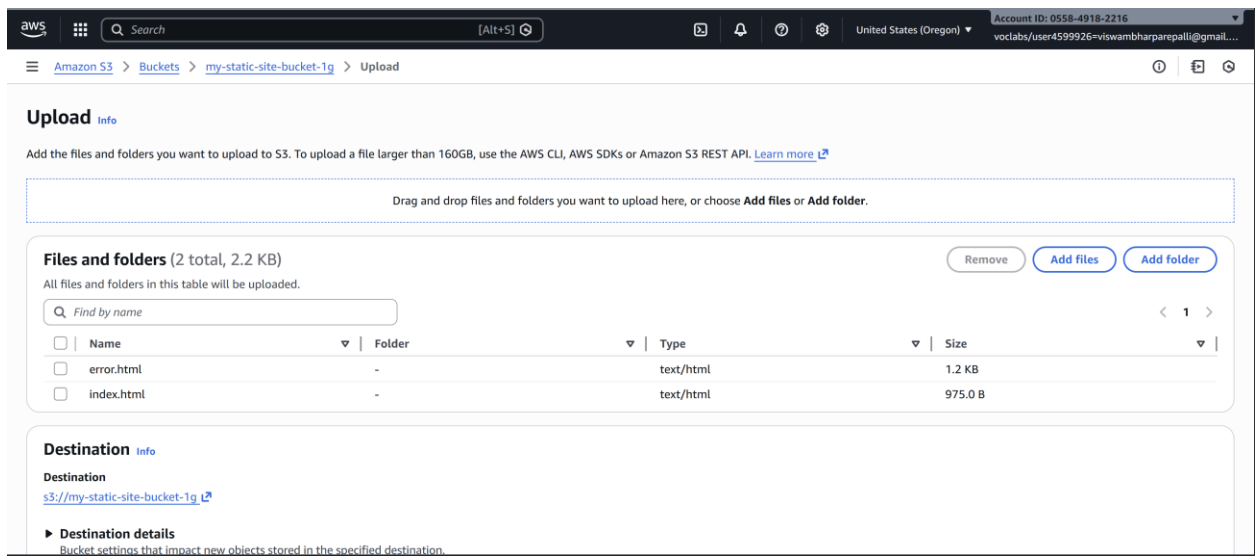
1. Login to AWS Management Console
2. Go to Services → S3
3. Click Create bucket
4. Enter:
 - Bucket name: my-static-site-bucket-123
 - Region: Nearest region
5. Under Block Public Access:
 - ✗ Uncheck Block all public access
 - ✓ Acknowledge the warning
6. Click Create bucket

✓ Bucket created successfully

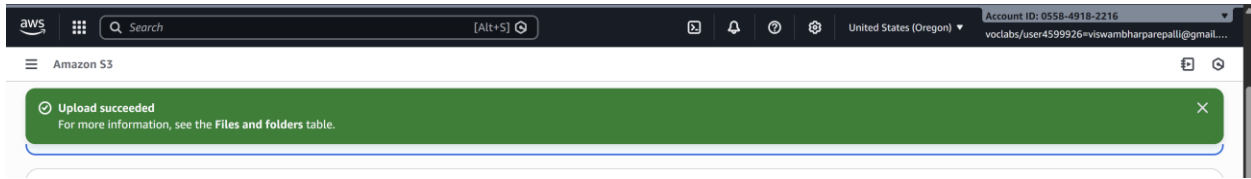


2 Upload Website Files

1. Open the bucket
2. Go to Objects tab
3. Click Upload
4. Click Add files
5. Select:
 - o index.html
 - o error.html
6. Click Upload

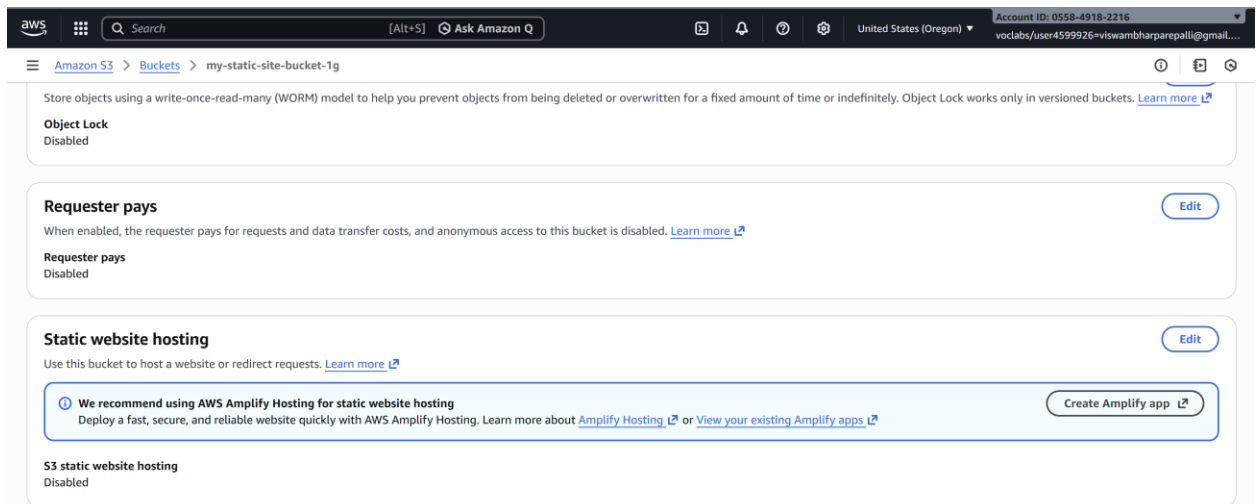


✓ Files uploaded

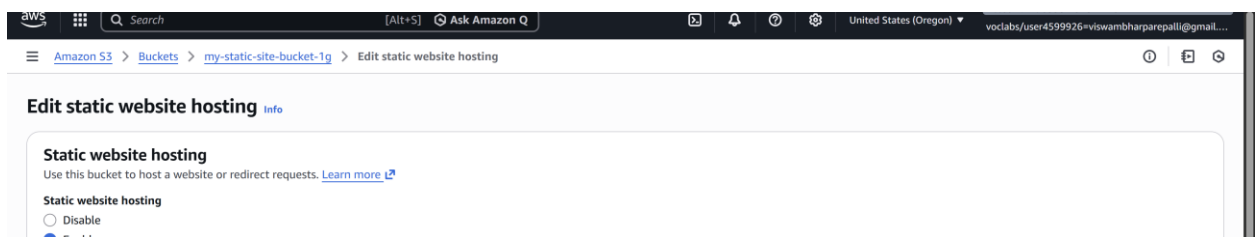


8. Enable Static Website Hosting

1. Go to Properties tab
2. Scroll to Static website hosting



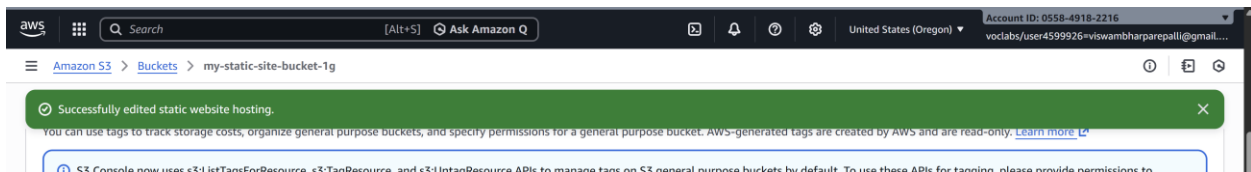
3. Click Edit
4. Select Enable



6. Choose Host a static website
7. Enter:
 - o Index document: index.html
 - o Error document: error.html

9. Click Save changes

✓ Static website hosting enabled



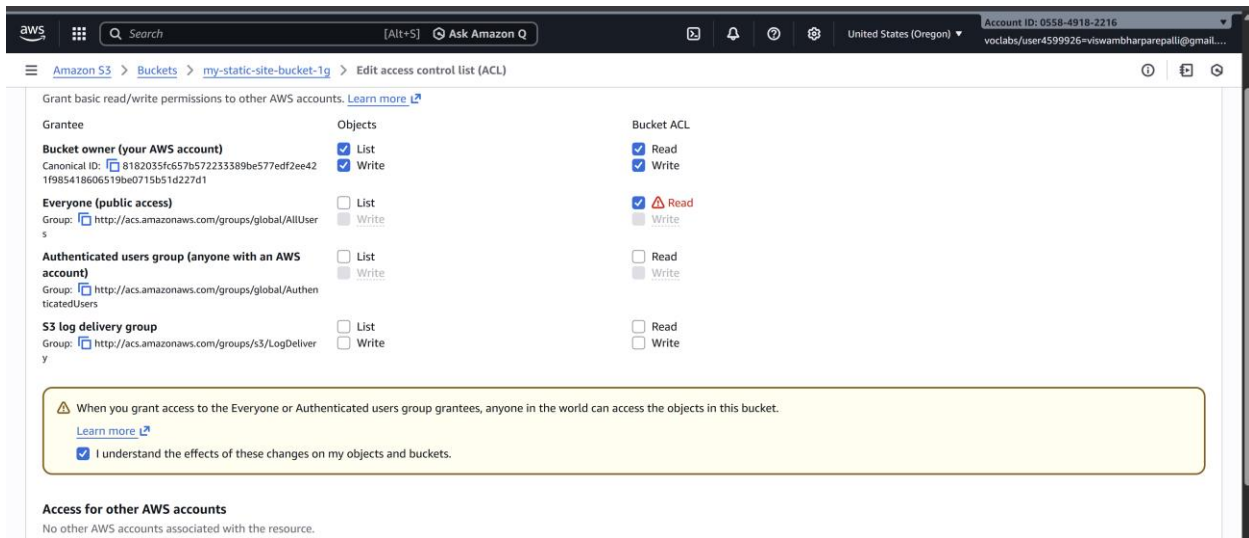
4. Enable Bucket-Level Public Access (ACL)

◆ Purpose

Allows public users to access objects inside the bucket.

1. Go to Permissions tab
2. Scroll to Access Control List (ACL)
3. Click Edit
4. Under Public access:
 - Everyone (public access) → ✓ Read
5. Click Save changes

✓ Bucket-level public read access enabled



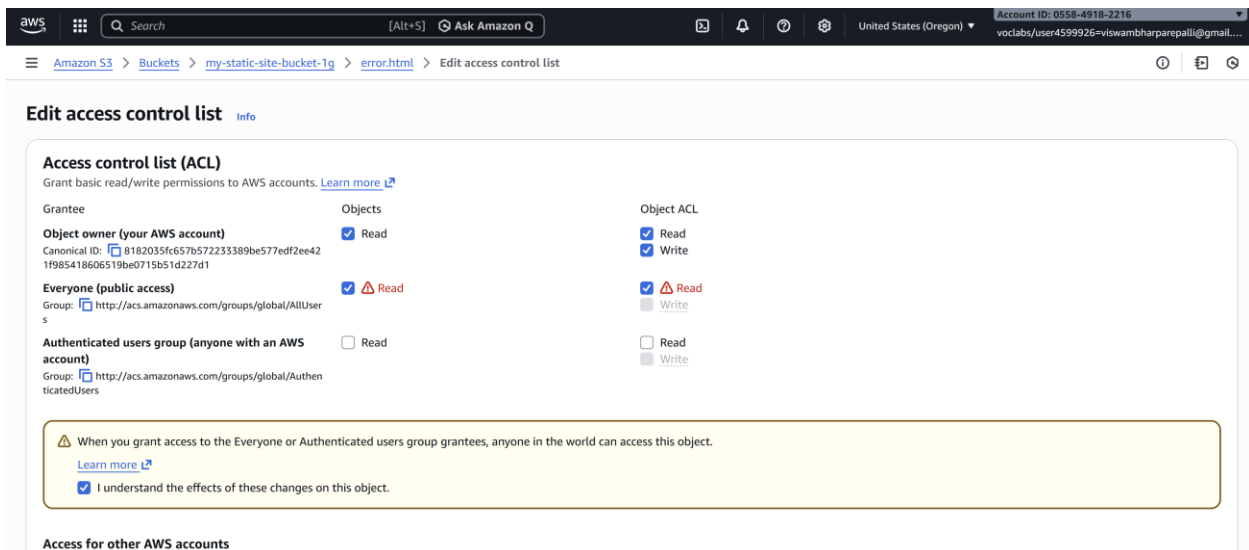
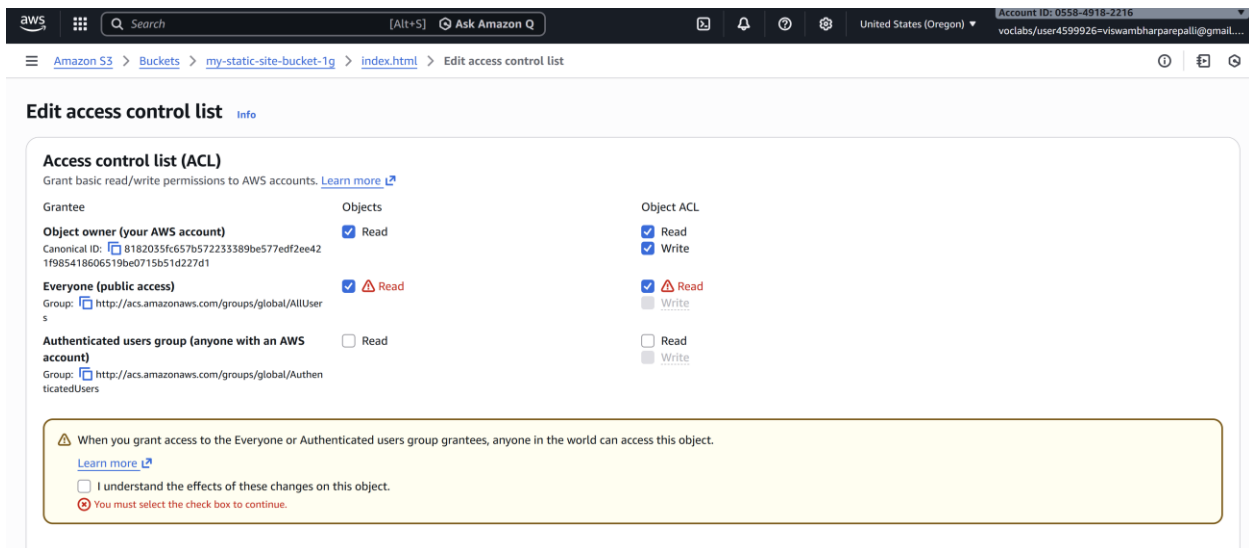
🔗 Enable Object-Level Public Access (ACL)

Steps (For Each File)

1. Go to Objects tab
2. Click on index.html
3. Go to Permissions
4. Scroll to Access Control List (ACL)
5. Click Edit
6. Under Public access:
 - Everyone (public access) → ✓ Read
7. Save changes

🔄 Repeat the same steps for error.html

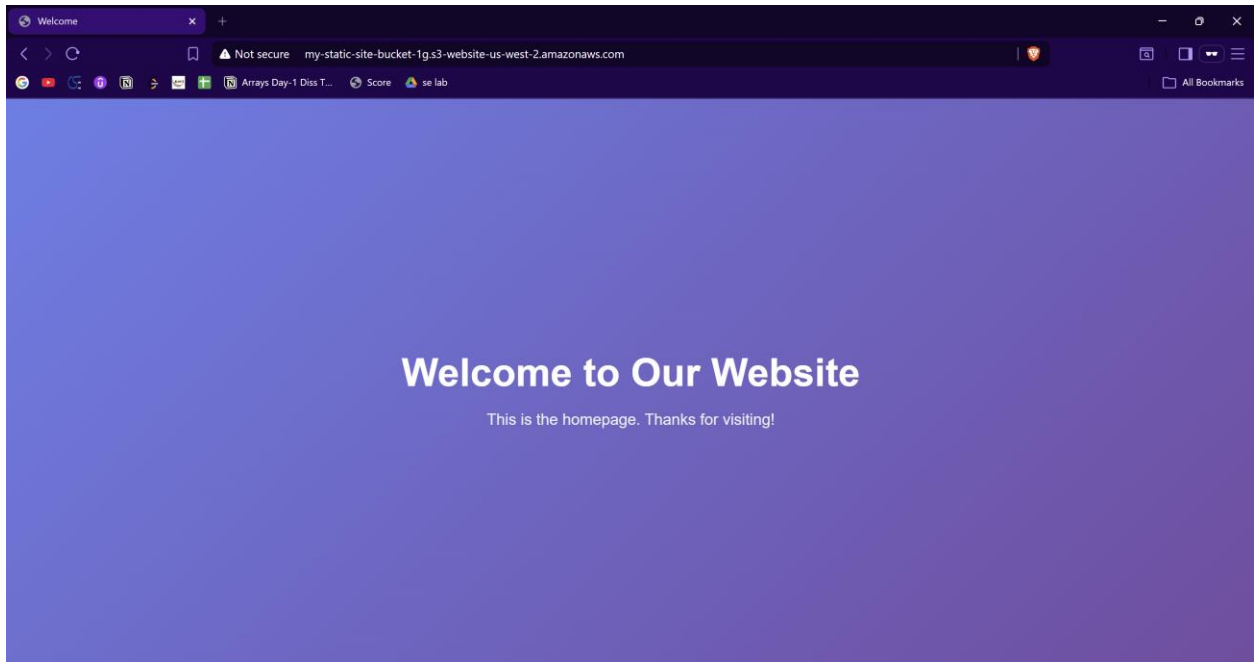
✓ Object-level public access enabled



Access the Static Website

1. Go to Properties
2. Scroll to Static website hosting
3. Copy the Bucket website endpoint
4. `http://my-static-site-bucket-123.s3-website-region.amazonaws.com`
5. Paste into browser

✓ index.html page loads successfully



❗ Error Page Output Scenario

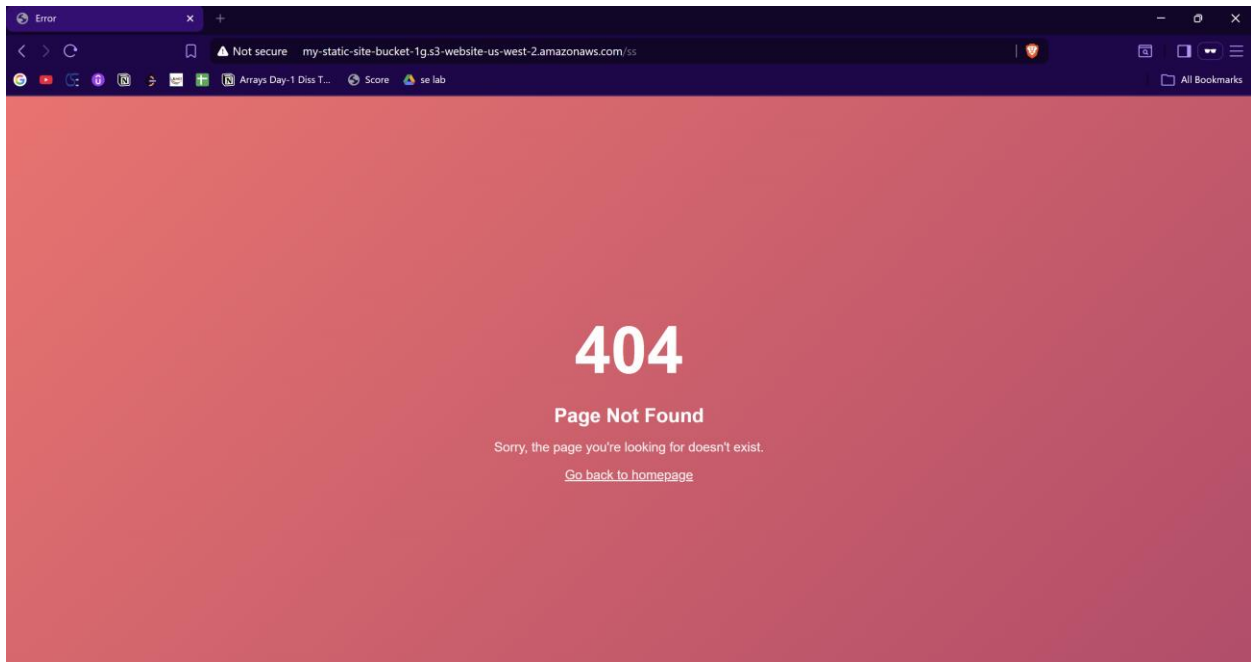
💎 Test Error Page

- Open an invalid URL:
- <http://bucket-name.s3-website-region.amazonaws.com/invalid.html>

🔗 Output:

- error.html page is displayed

✓ Error page working correctly



Scenario-Based Questions

1. A developer uploads an object to an S3 bucket but cannot access it from the browser, even though the bucket exists. What permission-related issue could be the cause?

Answer: Most likely the object isn't public. Either the object ACL is private or Block Public Access is on, so even though it's uploaded, the browser can't read it.

2. An S3 bucket policy allows public read access, but a specific object inside the bucket is still not accessible. Why might this happen?

Answer: Because object-level permissions can override bucket policy. If that specific object is private (ACL), it won't be accessible even if the bucket is public.

3. A bucket has **no bucket policy**, but individual objects have public ACLs. Will users be able to access those objects? What determines the final access decision?

Answer: Yes — *unless* Block Public Access is enabled. Final access is decided by the combo of bucket policy + object ACL + Block Public Access.

4. A company wants to ensure **no public access** to any object in an S3 bucket, even if someone mistakenly applies a public ACL. What S3 feature should be enabled?

Answer: Enable **S3 Block Public Access (at bucket or account level)**. This overrides any accidental public ACLs or policies. Hard stop.

5. You attempt to create an S3 bucket named my-company-data but receive a naming error. What S3 bucket naming rules might have been violated?

Answer: Bucket names must be globally unique, lowercase, no underscores, 3–63 chars, and look like DNS names. You probably violated one of those.

6. Two different AWS accounts try to create a bucket named project-backup-2026. One succeeds, the other fails. Why?

Answer: S3 bucket names are **global**. First account wins — second fails because the name already exists worldwide.

7. You enabled **static website hosting** on an S3 bucket and uploaded files, but accessing the website URL returns **403 Forbidden**. What is the most likely cause?

Answer: Classic mistake: objects aren't public. Static website hosting needs public read access on the files, not just the bucket.

8. Static website hosting is enabled, but accessing the root URL shows an error saying **index.html not found**. What is the actual issue?

Answer: You either didn't upload index.html, or you didn't configure it as the index document in website settings.

9. What HTTP error is returned when the **index document is missing** in S3 static website hosting, and why does this happen?

Answer: You usually get **403 Forbidden** — because *S3 can't find or access* the index file, so it blocks the request.

10. A bucket policy allows public access, but **static website hosting still doesn't work**. What object-level setting must also be verified?

Answer: Check the **object ACL**. Every website file (HTML/CSS/JS) must allow public read, not just the bucket.

11. You accidentally overwrite an important object in S3. Versioning was enabled earlier. How can the original file be recovered?

Answer: Easy fix: go to the object's versions and restore the previous version. That's literally why versioning exists.

12. A user deletes an object from a versioned S3 bucket. Is the data permanently removed? Explain what actually happens.

Answer: Nope only a *delete marker* is added. The older versions still exist and can be recovered anytime.

13. You enabled Cross-Region Replication (CRR) on a bucket, but objects are not replicating. What configuration might be missing?

Answer: You probably forgot to enable **versioning on BOTH source and destination buckets** — CRR won't work without it.

14. After enabling CRR, only newly uploaded objects are replicated, but old objects are not. Why?

Answer: CRR only applies to **new objects** by default. Existing data must be replicated manually (via batch replication).

15. A company wants to replicate S3 data to another region for **disaster recovery and compliance**. What are the key prerequisites for enabling CRR?

Answer: Versioning on both buckets, replication IAM role, destination bucket, and CRR rules — miss any one and it won't work.
